TWO DUALIZING COMPLEXES DIFFER BY TENSORING WITH AN INVERTIBLE OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-dualizing/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Allowed standard prerequisites (editorial).

- The property of dualizing complexes that identifies the double dual of the structure sheaf with the other dualizing complex up to an invertible object. Invoked at the start of the counted proof (source lines 2928-2930). Produces the invertible object that the statement predicts.
- The technical hom-evaluation isomorphism for complexes of finite tor dimension. Invoked inside the counted proof (source lines 2936-2939). Converts the invertibility into the tensor description of the internal hom between the two dualizing complexes.

Source reference. Original lemma lemma-dualizing: sources/stacks-dualizing/pinned-original-source.tex, source lines 2818–2818. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-equivalence-comes-from-invertible: sources/stacks-dualizing/pinned-original-source.tex, source lines 2859–2859. Navigation pointer only: the counted text refers to the Stacks item lemma-equivalence-comes-from-invertible, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-internal-hom-evaluate-isomorphism-technical: sources/stacks-dualizing/pinned-original-source.tex, source lines 2836–2836. Navigation pointer only: the counted text refers to the Stacks

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item more-algebra-lemma-internal-hom-evaluate-isomorphism-technical, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring. If $\omega_A^\bullet$ and $(\omega'_A)^\bullet$ are dualizing complexes, then $(\omega'_A)^\bullet$ is quasi-isomorphic to $\omega_A^\bullet \otimes_A^{\mathbf{L}} L$ for some invertible object L of $D(A)$.*

Proof. By Lemmas lemma-dualizing and lemma-equivalence-comes-from-invertible the functor $K \mapsto R\mathrm{Hom}_A(R\mathrm{Hom}_A(K, \omega_A^\bullet), (\omega'_A)^\bullet)$ maps A to an invertible object L. In other words, there is an isomorphism

$$L \longrightarrow R\mathrm{Hom}_A(\omega_A^\bullet, (\omega'_A)^\bullet)$$

Since L has finite tor dimension, this means that we can apply More on Algebra, Lemma more-algebra-lemma-internal-hom-evaluate-isomorphism-technical to see that

$$R\mathrm{Hom}_A(\omega_A^\bullet, (\omega'_A)^\bullet) \otimes_A^{\mathbf{L}} K \longrightarrow R\mathrm{Hom}_A(R\mathrm{Hom}_A(K, \omega_A^\bullet), (\omega'_A)^\bullet)$$

is an isomorphism for K in $D^b_{Coh}(A)$. In particular, setting $K = \omega_A^\bullet$ finishes the proof. □